TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

STUDENT NAME:			
STUDENT ID NUMBER:		TION NUMBER IN THIS COURSE:	

Homework1 – SPRING 2024

COURSE NAME:	*Computer Architecture*
COURSE NUMBER:	SOC2060
EXAMINATION DATE:	TIME:
EXAMINATION DURATION:	
ADDITIONAL MATERIALS ALLOWED TO USE:	None
SPECIAL INSTRUCTIONS:	*- Answers will only be evaluated if they are readable.* *- Show all your work. For some questions, you may get partial credit even if the end result is wrong due to a calculation mistake. If you make assumptions, state your assumptions clearly and precisely.*

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write your Name (*capital letters*), ID number and Section number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

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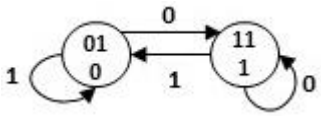
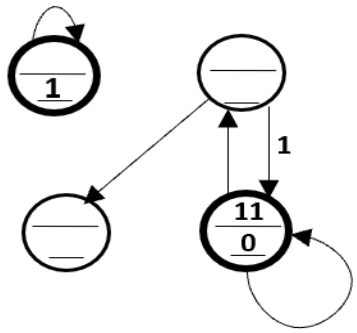
1.[12 points] Finite State Machine

(a)[12 points] Given the partially completed truth table and FSM diagram below, complete all the missing entries in the truth table and the FSM diagram. In each state circle, the top entry is S_1S_0 and the bottom entry is the value of Out. Make sure that you have labeled all missing states, inputs and outputs, and that you have added and labeled any missing transitions in the FSM.



Note that in the above diagram, '01' and '11' in the circle are encoded states, and '0' and '1' in the circle are outputs. In state '01', if input is '0', next state is '11' and output is '0'. In state '01', if input is '1', next state is '01' and output is '0'.

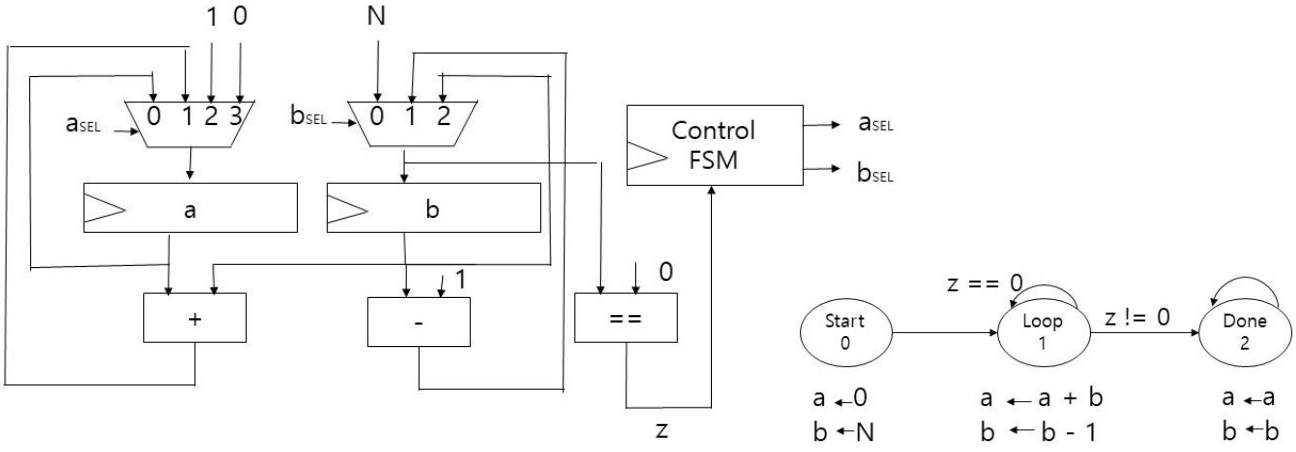


S_1	S_0	In	S_1'	S_0'	Out
0	0	0	1	0	0
0	0	1	1	1	0
0	1	0	0	0	1
0	1	1			
1	0	0			
1	0	1	0	1	
1	1	0			0
1	1	1	0	1	

2.[29 points] Programmable Machine

(a)[9 points] Below is the picture of a small processor and a FSM which computes

F = N + (N-1) + (N-2) + … + 2 + 1



Complete following tables

s	z	aSEL	bSEL	s'
00	0			
00	1			

s	z	aSEL	bSEL	s'
01	0			
01	1			

s	z	aSEL	bSEL	s'
10	0			
10	1			

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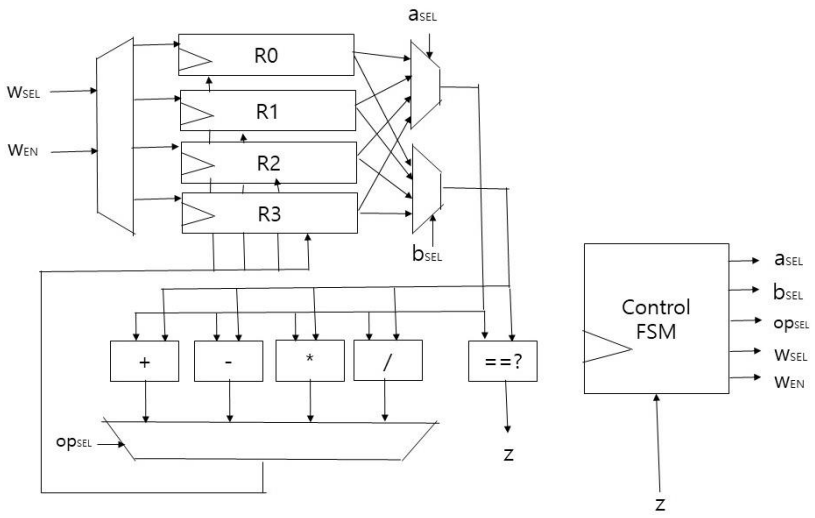
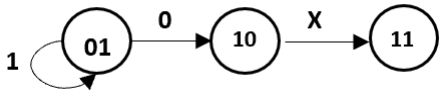
(b)[20 points] Below is the picture of a programmable machine



Initially, R0 contains N, R1 contains k, R2 contains -1, R3 contains 1. In the final state, R3 has final result. Finish the FSM to calculate N ** K (exponential). Start state is '00' and final state is '11'.

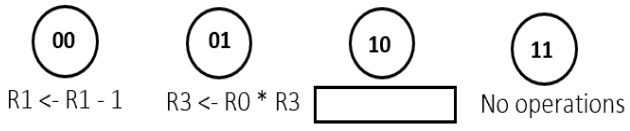
1. Write the operation of state '01' inside the rectangle.
2. Draw arrows, and write 'z' value (0 or 1) above each arrow. If 'z' value can be either 0 or 1, write 'X' above the arrow.



In the state '01' of the above example FSM, if z == 0, next state is '10', otherwise, next state is '01'.

In the state '10' of the above example FSM, next state is '11' regardless of the value of 'z'.

3. Finish following tables



s	z	opSEL	WSEL	WEN	aSEL	bSEL	s'
00	0						
00	1						

s	z	opSEL	WSEL	WEN	aSEL	bSEL	s'
01	0						
01	1						

s	z	opSEL	WSEL	WEN	aSEL	bSEL	s'
10	0						
10	1						

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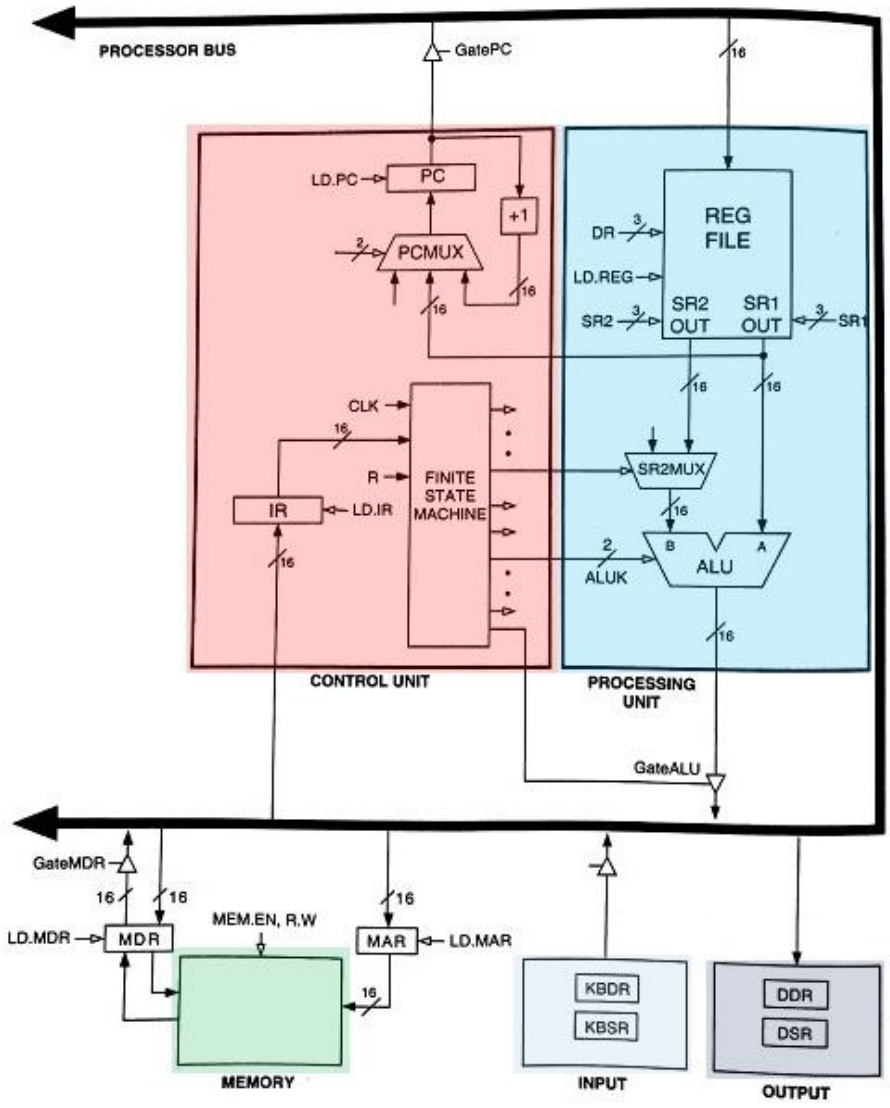
3.[9 points] LC-3 Architecture

Below is the picture of LC-3 processor



(a)[3 points] Fill in the values of control signals in the first instruction cycle? In that cycle, the following operations are processed at the same time.

MAR <- PC

PC <- PC + 1

GatePC	GateMDR	LD.REG	LD.IR	LD.PC	LD.MAR	LD.MDR

(b)[3 points] Fill in the values of control signals in the STORE RESULT cycle? The instruction is R3<-R2 + R1.

GateALU	GateMDR	LD.REG	LD.IR	LD.PC	LD.MAR	DR

(c)[3 points] Fill in the values of control signals in the FETCH OPERAND cycle? The instruction is LDR R1 R2 #0

GateALU	GateMDR	LD.REG	LD.IR	LD.PC	LD.MAR	LD.MDR

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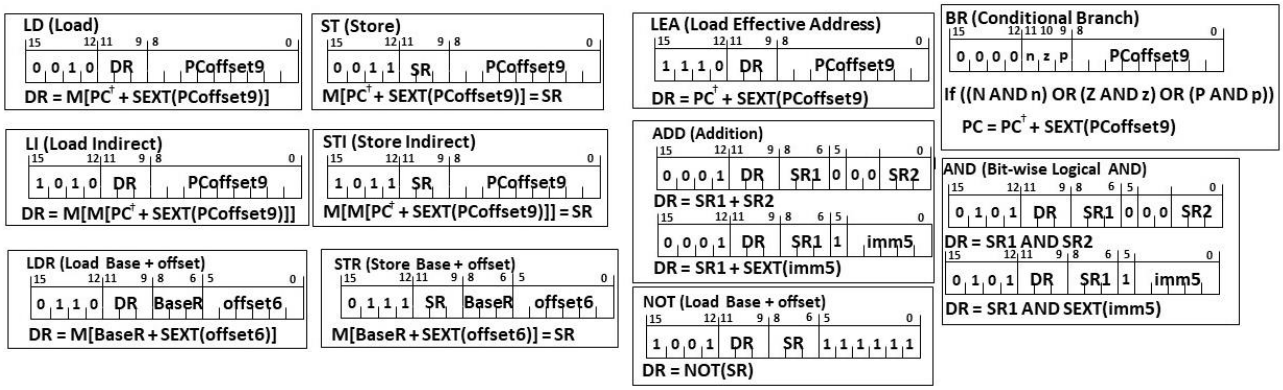
4.[24 points] LC-3 Machine Language



(a)[8 points] Finish LC-3 machine code to load R1 with integer 224 (R1 <- 224). Assume that your memory and registers are empty, and try to make it run as fast as possible. Write semantics of your instructions to the right of your instructions.

0	1	0	1	0	0	1	0	0	1	1	0	0	0	0	0	R1 <- 0

(b)[8 points] Now, R1 has integer 224 and integer k is stored at location x30FF in memory. Finish LC-3 machine code to calculate 224*k and save the result to R0. Write semantics of your instructions to the right of your instructions. Your instructions start at location x3000.

x3000	0	1	0	1	0	1	1	0	1	1	1	0	0	0	0	0	R3 <- 0
x3001																	R2 <- M[x30FF]
x3002																	If (R2==0) Branch out of loop
x3003																	
x3004																	
x3005																	Branch to start of loop
x3006																	R0 <- R3 + 0

(c)[8 points] Above code may be inefficient when integer k is bigger than 224. Write machine code to make it more efficient by comparing 224 with k. Later you're going to insert this machine code between instructions at x3001 and x3002 without updating your code in problem **(b)**

x3002																	R4 <- R1 – R2 (Use R5 to store the 2's complement of R2)
x3003																	
x3004																	
x3005																	If (R4 >= 0) Branch to x3009
X3006																	Swap R1 and R2 (Use R5 for storing the temporary value)
X3007																	
x3008																	
x3009																	R2 <- R2 + 0

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5.[16 points] LC-3 Machine Language

Address	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x30F6	1	1	1	0	0	0	1	1	1	1	1	1	1	1	0	1
x30F7	0	0	0	1	0	1	0	0	0	1	1	0	1	1	1	0
x30F8	0	0	1	1	0	1	0	1	1	1	1	1	1	0	1	1
x30F9	0	1	0	1	0	1	0	0	1	0	1	0	0	0	0	0
x30FA	0	0	0	1	0	1	0	0	1	0	1	0	0	1	0	1
x30FB	0	1	1	1	0	1	0	0	0	1	0	0	1	1	1	0
x30FC	1	0	1	0	0	1	1	1	1	1	1	1	0	1	1	1

(a)[4 points] The PC has just become x30F8 at the Fetch phase. What is the value of R1?

R1 = _____

(b)[4 points] The PC has just become x30F9 at the Fetch phase. What is the value of R2?

R2 = _____

(c)[4 points] The PC has just become x30FB at the Fetch phase. What is the value of R2?

R2 = _____

(d)[4 points] The PC has just become x30FC at the Fetch phase. What is the value of R2?

R2 = _____

6.[10 points] ISA and ISA Trade-offs

(a)[10 points] Circle Uarch or ISA corresponding to whether each of the following characteristics is better classified as "microarchitecture" or "ISA":

Size of the general purpose registers: Uarch / ISA

Load/store vs. memory/memory: Uarch / ISA

Support for virtual memory: Uarch / ISA

The number of ALU's in the CPU: Uarch / ISA

The number of general purpose registers in the CPU: Uarch / ISA

Variable length instructions: Uarch / ISA

Exception and interrupt handling: Uarch / ISA

Access protection: Uarch / ISA

Privilege modes: Uarch / ISA

Special I/O instructions: Uarch / ISA